

DataCORE

Reproducibility, Slide Sync & AI Attribution

Afternoon: DataCORE

1. **Reproducibility, Open Science & Your Code Portfolio (1:00–1:45)**
2. **Presentation Data Storytelling — Slide Sync (1:45–2:45)**
3. **Scientific Storytelling — AI Attribution Statement (2:45–3:00)**

A finding only counts if someone else can rebuild it. Today you make your notebook shareable, sync it to your slides, and disclose your AI use.

1:00 – 1:45 · Reproducibility

A finding only matters if it replicates

Open science means sharing your **data and code**, not just your conclusions.



Others can **check** your work — and build on it.



Transparency is a **defense**: nothing to hide.



Reproducible work is **reusable** work.

 **The challenges are real:** privacy of participants, credit and being scooped, and the plain effort of cleaning a notebook enough for a stranger to follow. Worth it anyway.

 **Today's goal:** clean your analysis notebook so another researcher could open it and understand it without you in the room.

Name things like you mean it

Before

After

```
df2 = pd.read_csv("data.csv")      # what is df2?  
x = df2[df2.a > 0]                 # what is a?  
y = x.groupby("g").b.mean()        # g? b?
```

 A variable name should tell a reader **what it holds** without them running the cell. `df2` costs a stranger five minutes; `health_outcomes_clean` costs them nothing.

Cut the clutter

Before you share, delete the scaffolding you built the analysis on:



Dead code — cells that no longer run or feed anything.



Redundant cells — the same thing tried five ways.



Debug **print()** spam you left behind.



The test: *Restart & Run All*, top to bottom, with no errors and no leftovers. If a cell isn't needed for that clean run, it shouldn't ship.



Keep one **comment per step of intent** — enough to follow, not a novel.

Two Markdown cells that frame the notebook

Top of the notebook — a header cell:

Wage & Experience Analysis

Authors: Team 3

Date: 2026-07-14

Data source: 2023 Wage Survey (N = 220)

Research question: Does hourly wage
rise with years of experience?

Bottom of the notebook — a limitations cell:

Limitations

- Observational data – no causal claims
- Wages self-reported; possible recall error
- One survey year – may not generalize

 The header answers *"what is this and who made it?"* in five seconds; the limitations cell answers *"what shouldn't I conclude from it?"* Together they make the notebook self-contained.

AI assist – a reader's-eye review

Ask your AI tool:

"Review my notebook and suggest three ways to make it easier for another researcher to understand."

Good suggestions to expect:

- Rename cryptic variables
- Add a comment above each analysis step
- Split a giant cell into logical pieces

 **You decide** which suggestions to take – and note that you used AI for this pass (we'll write that statement at 2:45).

45:00

1:45 – 2:45 · Data Storytelling

Every chart on a slide = the chart in your notebook

Open your **slides** and your **analysis notebook** side by side. They must tell the **same** story, chart for chart.

A chart that appears on a slide but not in the notebook — or looks different from it — is a **red flag** to any careful audience.

 If a reviewer can't find the slide's chart in your notebook, they can't trust it — and they'll wonder what else doesn't match.

☑️☰ The four Slide-Sync checks

- 1 **Chart title matches the slide text** — the same claim in both places.
- 2 **Axis labels legible at slide zoom** — minimum **14pt** font.
- 3 **A caption sits below each chart** — what it shows, the finding, the source.
- 4 **No orphan charts** — every chart is explained in the speaker notes.

👤👤 **Peer check:** swap laptops with another team. Can they say what each chart shows **without asking you?** If not — revise until they can.

60:00

What "presentation-ready" looks like



✓ **Title is the finding**, not "wage vs experience".

✓ **Axis labels ≥ 14 pt**, readable from the back.

“ *Caption:* "Hourly wage against years of experience for 220 workers; wages rise roughly \$1 per year. Source: 2023 Wage Survey."

2:45 – 3:00 · Scientific Storytelling

Write your AI attribution statement

Fill in the brackets for **your** project, then add it to your presentation:

*"AI tools were used in this project for **[specific tasks – e.g. code drafting, language editing]**. All analyses were conducted and verified by the research team. All interpretations and conclusions are our own."*

 This mirrors the transparency standards now required by major journals – **Nature, Science, APA.**

 **Practice saying it out loud** as part of your presentation – disclosure is spoken, not buried on a final slide.

15:00

